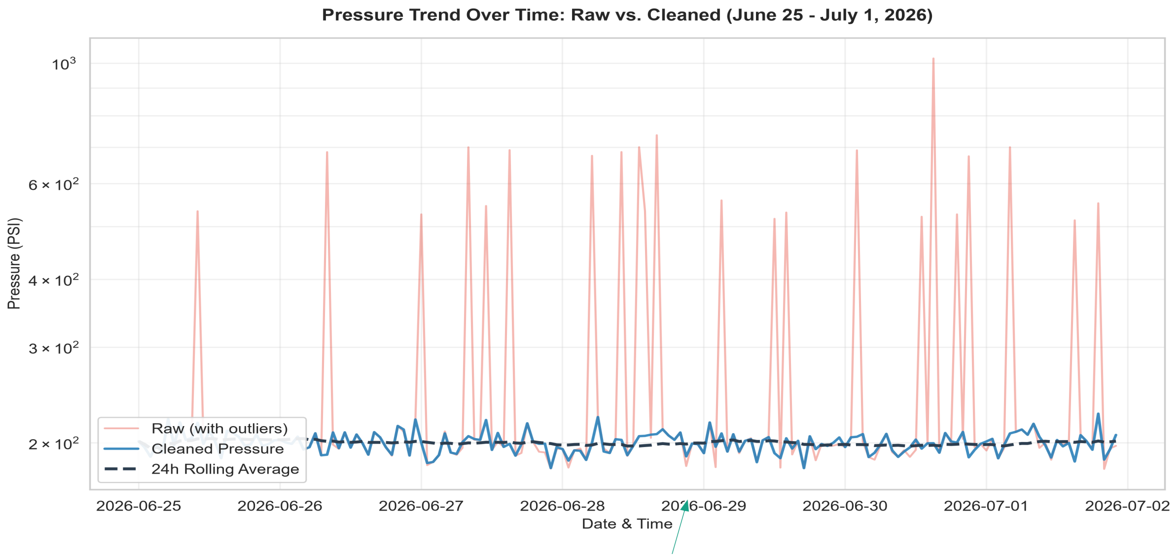
OPERATIONAL PROCESS INSIGHT REPORT

DATA WRANGLING & ANALYSIS | PREPARED FOR PLANT MANAGER | BY INUKA TECH

1. Context: Unmasking Corrupted Raw Operations Log

The raw plant sensor log was severely degraded by electronic malfunctions and physical dropouts, presenting an inaccurate picture of plant operations. A structural audit revealed critical data quality failures, including 15 duplicate entries and a swapped day-month timestamp typo on July 1st that skewed chronological trends. More critically, the dataset was littered with impossible readings: pressure sensors recorded massive spikes up to 15,000 PSI and dropouts of -50 PSI, while temperature logs showed spikes of 1,500°C and dropouts at absolute zero (-273.15°C) and exactly 0.0°C. These corrupted readings severely distorted the overall plant metrics—misrepresenting average pressure as a volatile 255.25 PSI instead of the actual ~200 PSI baseline—rendering the raw log useless for operational safety and planning.



2. Insight: Unveiling the Stable Baseline & Night-Shift Dip

Data cleaning and linear interpolation revealed that the plant is operating within a stable baseline, but exposed a subtle, recurring night-shift pressure dip. Removing the invalid spikes and dropouts shows that the plant's actual pressure resides consistently within a safe operating window of 120 to 280 PSI (averaging 200.13 PSI), while temperature is maintained between 45°C and 85°C. However, with the extreme sensor noise eliminated, the cleaned data exposes a distinct operational anomaly: a nightly pressure drop centered in **Zone West** during the Night shift, where average pressure drops to **192.42 PSI** (a 4% decrease compared to Day shifts). This nightly dip was completely hidden by the 15,000 PSI spikes in the raw log, but is now clearly visible as a systematic trend repeating across all seven days of the log.

3. Action: Recommended Corrective Maintenance & Upgrades

To prevent process degradation and ensure pressure stability, immediate inspection and sensor upgrades are required in Zone West. First, dispatch maintenance teams to inspect Zone West during the Night shift (10 PM - 6 AM) to locate potential pneumatic leaks, valve failures, or utility supply drops that explain the 192.42 PSI dip. Second, replace the aging pressure and temperature sensors in Zone West and Zone South, as they exhibited the highest frequency of signal dropouts and extreme spikes. Finally, configure the plant's Supervisory Control and Data Acquisition (SCADA) system to flag any night-shift pressure drops below 180 PSI, ensuring real-time alerting before pressure dips trigger automatic safety shutdowns.